

Assignment 1: Sales Prediction using Linear Regression

Objective:

The objective of this assignment is to apply Linear Regression to predict product sales using the BigMart dataset. Students will perform data preprocessing, exploratory data analysis (EDA), feature engineering, and build a regression model to understand how different factors influence sales.

Dataset:

Big Mart Sales Train Dataset:

https://raw.githubusercontent.com/ammishra08/Machine-Learning/refs/heads/master/Datasets/bigmart_train.csv

Big Mart Sales Test Dataset:

https://raw.githubusercontent.com/ammishra08/Machine-Learning/refs/heads/master/Datasets/bigmart_test.csv

Problem Statement:

BigMart is a retail company with multiple outlets across different cities. The management wants to understand which factors influence product sales in different stores so that they can make better business decisions regarding inventory management, pricing, and marketing strategies.

You are given historical sales data containing information about products, store characteristics, and sales figures. Using this dataset, your task is to build a Linear Regression model to predict the sales of products in different BigMart outlets.

The dataset contains attributes such as product weight, product type, product visibility, outlet size, outlet type, location tier, and other relevant features that may affect sales.

You need to analyze the dataset and develop a Linear Regression model to estimate the target variable *Item_Outlet_Sales*.

Assignment Tasks:

1. Load the dataset into a Pandas DataFrame and display the first 10 records.
2. Identify the number of rows, columns, and data types of each feature in the dataset.
3. Detect and report missing values present in the dataset.
4. Perform data cleaning by handling missing values appropriately.
5. Separate numerical and categorical variables in the dataset.
6. Perform exploratory data analysis to understand the distribution of numerical variables.
7. Analyze the relationship between independent variables and the target variable *Item_Outlet_Sales*.
8. Visualize correlations between numerical features using a correlation matrix.
9. Encode categorical variables using an appropriate encoding technique.
10. Split the dataset into training and testing sets.
11. Define independent variables (X) and the target variable (y).
12. Build a *Linear Regression* model using the training dataset.

13. Use the trained model to predict sales for the test dataset.
14. Evaluate the model performance using R^2 Score, MAE , MSE , and $RMSE$.
15. Interpret the coefficients of the Linear Regression model to understand feature impact on sales.
16. Summarize the key insights obtained from the analysis and model results.